



## **Business Template**

# **BUYEVERYTHING GLOBAL RETAIL**

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Logo / Image

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# 1 BUSINESS DESCRIPTION

## 1.1 BUSINESS BACKGROUND

BuyEverything Global Retail is a volume omnichannel e-commerce platform distributing consumer goods across both domestic US and international markets. In a highly saturated and fast-paced retail ecosystem, operational success depends heavily on a deep understanding of customer buying behavior, supply chain efficiency, and cross-border commercial dynamics. To remain competitive, optimize pricing structures, and minimize overhead costs, the company aggregates huge volumes of raw transactional data from daily digital sales. Capturing this data across separate regional jurisdictions allows the business to track customer demand and financial viability on a global scale of all it can be done by collecting product sales information and analyzing the one using special tools.

## 1.2 PROBLEMS BECAUSE OF POOR DATA MANAGEMENT

Without a structured data architecture, the company faces severe data fragmentation silos across its regional operations. The US database tracks metrics using regional state structures, while the International database utilizes separate country-level identifiers with completely different column schemas and naming formats. Poor data governance creates immediate business challenges: incompatible Metrics: variations in field naming conventions, blind spots in profitability

## 1.3 BENEFITS FROM IMPLEMENTING A DATA WAREHOUSE

Implementing a centralized Data Warehouse solves structural errors by transforming messy, denormalized source files into a unified single source of truth.

Examples of analytical questions the Data Warehouse will answer:

- What is the true net profitability of specific categories once discounts, shipping overhead, and regional tax obligations are deducted?
- Which customer segments (e.g., Loyal vs. New) exhibit the highest Customer Lifetime Value (CLV) across different geographic areas?
- Return Pattern Analysis: Correlates high product return frequencies (Returned = Yes) with specific device types, regions, or product subcategories to eliminate defective suppliers

## 1.4 DATASETS DESCRIPTION

The first dataset **US Domestic Operations** contains information about sales on the US market.

The second dataset **International Operations** contains information about sales on the international market. Both datasets contain similar attributes (name difference is provided in the table below)

International Operations does not contain information about customer gender and age

Location discrepancy - US contains State attribute and international dataset holds Country name

### Product Information:

- **PRODUCT\_ID**: The unique stock

- **CUSTOMER\_SEGMENT:** The specific loyalty or behavioral tracking cohort tier assigned to the consumer.

### Sales Representative Information

- **SALES\_REP\_ID:** Unique identification code assigned to the managing business sales representative.
- **SALES\_REP\_FIRST\_NAME:** The first name of the assigned corporate sales representative.
- **SALES\_REP\_LAST\_NAME:** The last name of the assigned corporate sales representative.
- **SALES\_REP\_EMAIL:** The internal corporate electronic mail address of the sales representative.

### Orders Information

- **ORDER\_ID:** Unique identification number assigned to each specific order or sales transaction.
- **ORDER\_DT:** The calendar date indicating when the customer transaction was created.
- **ORDER\_STATUS:** The execution status of the transaction fulfillment pipeline.
- **RETURNED** A binary flag field indicating whether the item was sent back by the consumer (Yes/No).
- **PAYMENT\_METHOD:** The settlement framework or bank asset type used by the customer.
- **UNIT\_PRICE:** The base listing price per single individual unit of the catalog item.
- **QUANTITY:** The total number of items or units purchased within this specific transaction.
- **DISCOUNT\_PERCENT:** The percentage deduction rate applied to the original base price of the item.
- **DISCOUNT\_AMOUNT:** The absolute currency value subtracted from the order total due to applied promotions.
- **SHIPPING\_COST:** The logistics and distribution transportation fee required to ship the order.
- **TAX\_AMOUNT:** The fiscal sales tax or local value-added tariff applied to the transaction invoice.
- **ORDER\_AMOUNT:** The final absolute monetary invoice total charged to the customer after all modifications.
- **COST\_AMOUNT:** The comprehensive calculated fulfillment expenditure required to complete the sale.
- **PROFIT\_MARGIN\_PERCENT:** The calculated net profitability ratio margin structured as a percentage.
- **PROFIT\_AMOUNT:** The net financial business revenue remaining after deducting costs from the order total

### Location

- **STATE/COUNTRY:** The primary macro-level geographical region identifier (US State name or global Sovereign Country name).
- **CITY :** The localized urban municipality or distribution terminal destination name within the target region.

### Sale channel

- **CHANNEL:** The high-level operational marketing and distribution mechanisms through which a consumer places an order.
- **CHANNEL\_NAME:** The digital platforms or storefront applications that map back to a parent channel category.

| US Domestic Operations | International Operations |
|------------------------|--------------------------|
|                        |                          |

| US Domestic Operations | International Operations  |
|------------------------|---------------------------|
| CUSTOMER_GENDER        |                           |
| CUSTOMER_SEGMENT       | CUSTOMER_STATUS           |
| SALES_REP_ID           | SALES_REP_ID_INT          |
| SALES_REP_FIRST_NAME   | SALES_REP_FIRST_NAME_INT  |
| SALES_REP_LAST_NAME    | SALES_REP_LAST_NAME_INT   |
| SALES_REP_EMAIL        | SALES_REP_EMAIL_INT_INT   |
| PRODUCT_ID             | PRODUCT_ID_INT            |
| PRODUCT_CATEGORY       | CATEGORY                  |
| PRODUCT_SUBCATEGORY    | SUBCATEGORY               |
| UNIT_PRICE             | PRICE_INT                 |
| QUANTITY               | QUANTITY_INT              |
| DISCOUNT_PERCENT       | DISCOUNT_PERCENT_INT      |
| DISCOUNT_AMOUNT        | DISCOUNT_AMOUNT_INT       |
| SHIPPING_COST          | SHIPPING_COST_INT         |
| TAX_AMOUNT             | TAX                       |
| ORDER_AMOUNT           | ORDER_AMOUNT_INT          |
| COST_AMOUNT            | COST_AMOUNT_INT           |
| PROFIT_MARGIN_PERCENT  | PROFIT_MARGIN_PERCENT_INT |
| PROFIT_AMOUNT          | PROFIT_AMOUNT_INT         |
| STATE                  |                           |
|                        | COUNTRY                   |
| CITY                   | CITY_INT                  |
| PAYMENT_METHOD         | PAYMENT                   |
| CHANNEL                | CHANNEL                   |
| CHANNEL_NAME           | CHANNEL_NAME              |

## Sales and Cost measures

### 1. Core Sales Measures (Revenue & Quantities)

These attributes capture the volume and gross financial inputs derived from customer purchasing transactions.

- **UNIT\_PRICE / PRICE:** The base, standard retail listing price per single individual unit of a catalog item before any promotional adjustments or regional taxes.
- **QUANTITY / QUANTITY:** The total integer count of item units purchased within a single transaction row, acting as the primary volume multiplier for total sales calculations.

- **ORDER\_AMOUNT / ORDER\_AMOUNT:** The final absolute gross monetary revenue invoiced to and paid by the customer. It represents the ultimate top-line sales figure for the transaction, calculated as:

## 2. Cost and Fulfillment Measures

These metrics capture the structural expenses incurred by the business to procure, ship, and settle the order.

- **COST\_AMOUNT / COST\_AMOUNT\_INT:** The total calculated base operational expenditure required to fulfill the transaction. This encompasses the wholesale Cost of Goods Sold (COGS), supply chain inventory procurement, manufacturing costs, or international landed tariffs.
- **SHIPPING\_COST / SHIPPING\_COST:** The logistics and freight distribution fee required to securely transport the order from the fulfillment warehouse terminal to the customer's specified delivery location.
- **TAX\_AMOUNT / TAX:** The unavoidable fiscal sales tax, value-added tax (VAT), or regional duty tariff applied directly to the transaction invoice based on the customer's geographic location.

## 3. Promotional Reductions

Measures used to track marketing markdown performance and revenue leaks.

- **DISCOUNT\_PERCENT / DISCOUNT:** The percentage rate deduction applied to the original base unit price as part of a promotional campaign or customer segment benefit.
- **DISCOUNT\_AMOUNT / DISCOUNT\_AMOUNT:** The total absolute currency value deducted from the gross order total due to applied promotions or coupons.

## 4. Net Profitability Measures (Derived Margin Performance)

The final bottom-line health indicators resulting from subtracting business costs from incoming sales amounts.

- **PROFIT\_AMOUNT / PROFIT\_AMOUNT:** The net financial business revenue remaining after deducting all product fulfillment costs from the top-line order amount.
- **PROFIT\_MARGIN\_PERCENT / PROFIT\_MARGIN\_PERCENT:** The structural efficiency ratio representing net profitability, calculated to show what percentage of the incoming sales revenue is captured as clear profit

## The hierarchies

These hierarchies based on the available dataset attributes and the business scenarios that can be supported by the model

- Category > Subcategory > Product
- Channel > Channel Name
- Country > State > City

## 1.5 GRAIN / DIM / FACT

Dim Description

### Step 1

The source datasets strictly contain single-item transactions; multi-item orders are not provided. Consequently, the order level serves as the most atomic level of granularity available. Because of this traditional order-level summary metrics (such as ORDER\_AMOUNT, SHIPPING\_COST, and TAX\_AMOUNT) may be included as additive measurements at this grain.

### Step 3

Identifying dimensional tables

#### DIM\_PRODUCT

| Column name         | Description                                                                       | Data Type |
|---------------------|-----------------------------------------------------------------------------------|-----------|
| PRODUCT_ID          | Surrogate primary key for product identification                                  | INT       |
| PRODUCT_SRC_ID      | Unique identification number assigned to each specific order or sales transaction | VARCHAR   |
| PRODUCT_CATEGORY    | The standardized broad structural master classification of the product category.  | VARCHAR   |
| PRODUCT_SUBCATEGORY | The targeted nested subcategory group mapping exact inventory details.            | VARCHAR   |

Example with filled data

| PRODUCT_ID | PRODUCT_SRC_ID | PRODUCT_CATEGORY | PRODUCT_SUBCATEGORY |
|------------|----------------|------------------|---------------------|
| 2          | PROD01695      | Electronics      | Laptop              |

#### DIM\_CUSTOMER

| Column name         | Description                                                                       | Data Type |
|---------------------|-----------------------------------------------------------------------------------|-----------|
| CUSTOMER_ID         | Surrogate primary key for customer identification                                 | INT       |
| CUSTOMER_SRC_ID     | Unique alphanumeric identification key allocated to an individual customer.       | VARCHAR   |
| CUSTOMER_FIRST_NAME | The legal first given name of the customer profile.                               | VARCHAR   |
| CUSTOMER_LAST_NAME  | The family surname of the customer profile.                                       | VARCHAR   |
| CUSTOMER_EMAIL      | The verified digital mail communication address belonging to the customer.        | VARCHAR   |
| CUSTOMER_AGE        | The chronological age of the customer recorded at the time of purchase.           | INT       |
| CUSTOMER_GENDER     | The identified gender profile of the consumer.                                    | VARCHAR   |
| CUSTOMER_SEGMENT    | The specific loyalty or behavioral tracking cohort tier assigned to the consumer. | VARCHAR   |

Example with filled data

| CUSTOMER_ID | CUSTOMER_SRC_ID | CUSTOMER_FIRST_NAME | CUSTOMER_LAST_NAME | CUSTOMER_EMAIL                  | CUSTOMER_AGE | CUSTOMER_GENDER | CUSTOMER_SEGMENT |
|-------------|-----------------|---------------------|--------------------|---------------------------------|--------------|-----------------|------------------|
| 3           | CINT047528      | Fabrice             | Simon              | fabrice.simon047528@hotmail.com | 45           | Male            | Premium          |

**DIM\_EMPLOYEE**

| Column name          | Description                                                                        | Data Type |
|----------------------|------------------------------------------------------------------------------------|-----------|
| SALES_REP_ID         | Surrogate primary key for sales representative identification                      | INT       |
| SALES_REP_SRC_ID     | Unique identification code assigned to the managing business sales representative. | VARCHAR   |
| SALES_REP_FIRST_NAME | The first name of the assigned corporate sales representative.                     | VARCHAR   |
| SALES_REP_LAST_NAME  | The last name of the assigned corporate sales representative.                      | VARCHAR   |
| SALES_REP_EMAIL      | The internal corporate electronic mail address of the sales representative.        | VARCHAR   |

Example with filled data

| SALES_REP_ID | SALES_REP_SRC_ID | SALES_REP_FIRST_NAME | SALES_REP_LAST_NAME | SALES_REP_EMAIL             |
|--------------|------------------|----------------------|---------------------|-----------------------------|
| 2            | RINT032          | Michael              | Turner              | michael.turner@novamart.com |

**DIM\_LOCATION**

| Column name | Description                                                                                          | Data Type |
|-------------|------------------------------------------------------------------------------------------------------|-----------|
| COUNTRY_ID  | Surrogate primary key for sales representative identification                                        | INT       |
| COUNTRY     | The global Sovereign Country name                                                                    | VARCHAR   |
| STATE       | The primary macro-level geographical region identifier (US State).                                   | VARCHAR   |
| CITY        | The localized urban municipality or distribution terminal destination name within the target region. | VARCHAR   |

Example with filled data

| COUNTRY_ID | COUNTRY | STATE | CITY  |
|------------|---------|-------|-------|
| 2          | Fance   | n/a   | Paris |



## DIM\_CHANNEL

| Column name  | Description                                                                                                | Data Type |
|--------------|------------------------------------------------------------------------------------------------------------|-----------|
| CHANNEL_ID   | Surrogate primary key for channel identification                                                           | INT       |
| CHANNEL      | The high-level operational marketing and distribution mechanisms through which a consumer places an order. | VARCHAR   |
| CHANNEL_NAME | The digital platforms or storefront applications that map back to a parent channel category.               | VARCHAR   |

Example with filled data

| CHANNEL_ID | CHANNEL    | CHANNEL_NAME |
|------------|------------|--------------|
| 1          | Mobile App | iOS App      |

## DIM\_DATE

| Column name | Description                                                             | Data Type |
|-------------|-------------------------------------------------------------------------|-----------|
| ORDER_DT    | The calendar date indicating when the customer transaction was created. | DATE      |

## DIM\_ORDER (JUNK DIMENSION OF UNRELATED ATTRIBUTES)

| Column name           | Description                                                                       | Data Type     |
|-----------------------|-----------------------------------------------------------------------------------|---------------|
| ORDER_ID              | Surrogate primary key for order identification                                    | INT           |
| ORDER_SRC_ID          | Unique identification number assigned to each specific order or sales transaction | VARCHAR       |
| ORDER_STATUS          | The execution status of the transaction fulfillment pipeline.                     | VARCHAR       |
| RETURNED              | Binary flag indicating whether the item was returned by the customer (Yes/No).    | VARCHAR       |
| PAYMENT_METHOD        | The settlement framework or bank asset type used by the customer.                 | VARCHAR       |
| UNIT_PRICE            | The base listing price per single individual unit of the catalog item.            | NUMERIC(10,2) |
| DISCOUNT_PERCENT      | The percentage deduction rate applied to the original base price of the item.     | INT           |
| PROFIT_MARGIN_PERCENT | The calculated net profitability ratio margin structured as a percentage.         | NUMERIC(10,2) |

Example with filled data

| ORDER_ID | ORDER_SRC_ID | ORDER_STATUS | RETURNED | PAYMENT_METHOD | UNIT_PRICE | DISCOUNT_PERCENT | PROFIT_MARGIN_PERCENT | COUNTRY       | STATE | CITY   |
|----------|--------------|--------------|----------|----------------|------------|------------------|-----------------------|---------------|-------|--------|
| 2        | OINT000001   | Delivered    | No       | Wallet         | 18.38      | 0                | 22.18                 | United States | Texas | Denver |

## Step 4

Identifying fact table

**FCT\_ORDER\_DD**

| Column name            | Description                                                                                | Data Type     |
|------------------------|--------------------------------------------------------------------------------------------|---------------|
| <b>ORDER_ID</b>        | Unique identification number assigned to each specific order or sales transaction          | VARCHAR       |
| <b>PRODUCT_ID</b>      | The unique stock-keeping unit (SKU) identifier assigned to a retail item                   | VARCHAR       |
| <b>SALES_REP_ID</b>    | Unique identification code assigned to the managing business sales representative.         | VARCHAR       |
| <b>CUSTOMER_ID</b>     | Unique alphanumeric identification key allocated to an individual customer                 | VARCHAR       |
| <b>QUANTITY</b>        | The total number of items or units purchased within this specific transaction              | INT           |
| <b>DISCOUNT_AMOUNT</b> | The absolute currency value subtracted from the order total due to applied promotions.     | NUMERIC(10,2) |
| <b>SHIPPING_COST</b>   | The logistics and distribution transportation fee required to ship the order               | NUMERIC(10,2) |
| <b>TAX_AMOUNT</b>      | The fiscal sales tax or local value-added tariff applied to the transaction invoice        | NUMERIC(10,2) |
| <b>ORDER_AMOUNT</b>    | The final absolute monetary invoice total charged to the customer after all modifications. | NUMERIC(10,2) |
| <b>COST_AMOUNT</b>     | The comprehensive calculated fulfillment expenditure required to complete the sale.        | NUMERIC(10,2) |
| <b>PROFIT_AMOUNT</b>   | The net financial business revenue remaining after deducting costs from the order total    | NUMERIC(10,2) |

Example with filled data

| ORDER_ID | PRODUCT_ID | SALES_REP_ID | CUSTOMER_ID | QUANTITY | DISCOUNT_AMOUNT | SHIPPING_COST | TAX_AMOUNT | ORDER_AMOUNT | COST_AMOUNT | PROFIT_AMOUNT |
|----------|------------|--------------|-------------|----------|-----------------|---------------|------------|--------------|-------------|---------------|
| 4        | 4          | 7            | 5           | 1        | 18.38           | 0             | 22.18      | 371.46       | 121.24      | 39.1          |

## 2 BUSINESS LAYER 3NF

- The entities that has transitive dependencies and were normalized:
  - Order > Customer
  - Order > Product>Subcategory>Category
  - Order > Employee
  - Order > Location>Country > State > City
  - Order > Channel>Channel >Subchannel
  - Customer > Customer Version (SCD2)
- Customer information was selected as the most suitable candidate for SCD Type 2 implementation because customer attributes may change over time while historical values remain important for analysis.
- In situations where source datasets do not provide values for required attributes, placeholder values such as 'N/A' will be loaded during the ETL process. Default constraints were intentionally not used, following the recommendations provided during the course.
- The source datasets do not provide identifiers for product categories and product subcategories. Therefore, PRODUCT\_CATEGORY\_NAME and PRODUCT\_SUBCATEGORY\_NAME serve as the business identifiers and source identifiers for those entities.
- The dataset contains one product per order. As a result, a separate order line table was not required, and product information can be directly associated with each order record.

## 3 BUSINESS LAYER DIMENSIONAL MODEL

## 4 LOGICAL SCHEME

## 5 DATA FLOW

## 6 FACT TABLE PARTITIONING STRATEGY